

Week 7: Systemd & Service Management (Part 1)

Date: April 20, 2026

Study Time: 5-8 hours total

GOAL

Understand what services are running and why

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 6: How User Space Starts - systemd

Video Learning

- ☐ LearnLinuxTV - Search: "Systemd Basics"
- ☐ Professor Messer - Search: "Linux+ Systemd"

Concepts

- ☐ What is systemd vs init
 - ☐ Units and unit files
 - ☐ Targets (like runlevels)
 - ☐ Service dependencies
 - ☐ journald for logging
-

LAB WORK (3-4 hours)

List and Understand Services

```
bash
```

```
# List all units
```

```
systemctl list-units
```

```
# List only services
```

```
systemctl list-units --type=service
```

```
# Show all available units (even inactive)
```

```
systemctl list-unit-files
```

```
# List failed services
```

```
systemctl --failed
```

Check Service Status

```
bash
```

```
# Check specific service
```

```
systemctl status sshd
```

```
systemctl status ssh # Ubuntu
```

```
# Is service active?
```

```
systemctl is-active sshd
```

```
# Is service enabled at boot?
```

```
systemctl is-enabled sshd
```

Manage Services

```
bash
```

```
# Start service
```

```
sudo systemctl start SERVICE_NAME
```

```
# Stop service
```

```
sudo systemctl stop SERVICE_NAME
```

```
# Restart service
```

```
sudo systemctl restart SERVICE_NAME
```

```
# Reload config without restart
```

```
sudo systemctl reload SERVICE_NAME
```

Boot Configuration

```
bash
```

```
# Enable service at boot
```

```
sudo systemctl enable SERVICE_NAME
```

```
# Disable service at boot
```

```
sudo systemctl disable SERVICE_NAME
```

```
# Enable and start immediately
```

```
sudo systemctl enable --now SERVICE_NAME
```

```
# Disable and stop immediately
```

```
sudo systemctl disable --now SERVICE_NAME
```

View Logs

```
bash
```

```
# View service logs
```

```
journalctl -u sshd
```

```
# Follow logs in real-time
```

```
journalctl -u sshd -f
```

```
# Last 50 lines
```

```
journalctl -u sshd -n 50
```

```
# Since today
```

```
journalctl -u sshd --since today
```

Lab Tasks

- ☐ List all running services on your laptop
- ☐ List all running services on Fedora server
- ☐ List all running services on Pi devices
- ☐ Disable unnecessary services (document which ones and why)
- ☐ Enable Gitea to start on boot: `sudo systemctl enable gitea`
- ☐ Practice starting/stopping services safely

1. `systemctl`
 2. `journalctl`
 3. `systemctl list-units`
 4. `systemctl enable`
 5. `systemctl disable`
 6. `systemctl status`
 7. `systemctl start/stop/restart`
 8. `systemctl is-active`
 9. `systemctl is-enabled`
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EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant: 1.3 Given a scenario, configure and verify network connection parameters

NOTES

Services disabled and why:

Services enabled and why:

Completed: ☐

QUICK REFERENCE

Common Services:

- `sshd` / `ssh` - SSH server
- `network` / `NetworkManager` - Networking
- `firewalld` / `ufw` - Firewall
- `rsyslog` - Logging
- `cron` - Scheduled tasks

Service States:

- active (running)
- inactive (dead)
- failed
- activating (starting)
- deactivating (stopping)

Previous Week: SSH Hardening

Next Week: Creating Custom Services